Part B: Instruction Extension Breakdown

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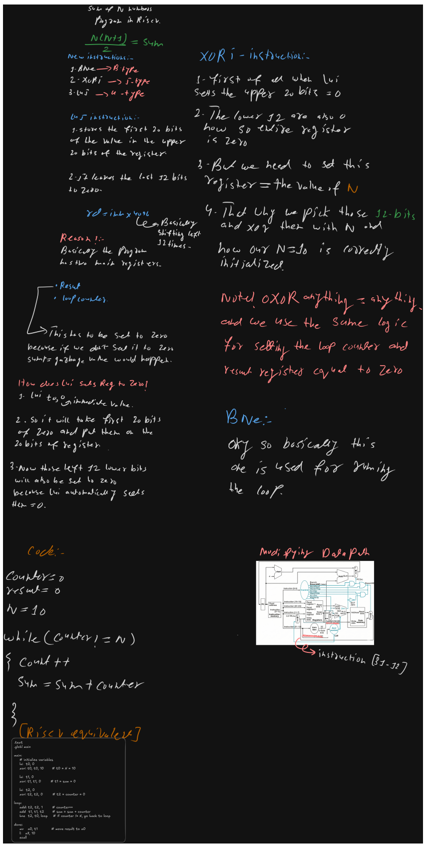
Brainstorming Notes



1. Instruction: LUI (U-Type)

Assembly: `lui x1, 0x12345`
Hex Code: `0x123450B7`
Binary: `00010010001101000101000010110111`

imm[31:12] (20 bits)	**rd** (5 bits)	**opcode** (7 bits)
00010010001101000101	00001	0110111
0x12345	x1	lui

2. Instruction: XORI (I-Type)

Assembly: `xori x2, x1, 6`
Hex Code: `0x0060C113`
Binary: `00000000011000001100000100010011`

imm[11:0] (12 bits)	**rs1** (5 bits)	**funct3** (3 bits)	**rd** (5 bits)	**opcode** (7 bits)
000000000110	00001	100	00010	0010011
6	x1	xor	x2	OP-IMM

3. Instruction: BNE (B-Type)

Assembly: `bne x4, x3, -4`
Hex Code: `0xFE321EE3`
Binary: `11111110001100100001111011100011`

<table>
<tr><th>imm[12]
(1 bit)</th><th>imm[10:5]
(6 bits)</th><th>rs2
(5 bits)</th><th>rs1
(5 bits)</th><th>funct3
(3 bits)</th><th>imm[4:1]
(4 bits)</th><th>imm[11]
(1 bit)</th><th>opcode
(7 bits)</th></tr>
<tr><td>1</td><td>111111</td><td>00011</td><td>00100</td><td>001</td><td>1110</td><td>1</td><td>1100011</td></tr>
<tr><td colspan="2" rowspan="2">imm = -4</td><td>x3</td><td>x4</td><td>bne</td><td colspan="2" rowspan="2">imm = -4</td><td>BRANCH</td></tr>
<tr><td colspan="3"></td><td></td></tr>
</table>

Immediate Calculation for BNE (-4):
Combined bits: `imm[12|11|10:5|4:1|0]`
Result: `1111111111100` (which is -4 in 13-bit two's complement).